

jayaprakash Makkena

✉ jayaprakash@usf.edu ☎ 8134940664 📍 Tampa, FL 🌐 jayaprakash makkena 🔄 jprakash120

SUMMARY

- Experienced Data Engineer with over two years of expertise in developing data pipelines, optimizing SQL performance, and automating ETL processes and data-driven decision-making.
- Adept at creating dynamic data visualizations using Power BI and Tableau, improving business decision-making through clear, actionable insights.
- I bring expertise in data analytics, machine learning, and tools like Python, SQL, and Power BI.
- Knowledge in Machine Learning, AI like RAG, LLM, TensorFlow, PyTorch, AI-Powered Analytics and predictive analytics to drive user targeting.
- Proven ability to mentor junior engineers, collaborate cross-functionally, and deliver high-quality results within fast-paced environments.

EDUCATION

Masters in Business Analytics and Information Systems,
University of South Florida, Tampa
GPA : 3.66/4.00

08/2023 – 05/2025

Bachelors in ECE, Vellore Institute Of Technology, Vellore

07/2018 – 05/2022

SKILLS

Programming Languages

Python, R, SQL, MATLAB,
PySpark, Pandas, Numpy, Java

Other Tools/Skills

Linux, Git, MSSuite, SAS, SPSS,
Predictive Analysis, Forecasting,
Dax, Critical thinking, Excel VBA,
Pivot Tables, VLOOKUP

Cloud/Bigdata

Microsoft Azure lake, Spark,
Hadoop, AWS

Databases

Oracle Database, Microsoft SQL
Server, MySQL, NoSQL, Oracle

Business Intelligence Tools

Tableau, Microsoft Power BI,
Google Looker Studio, Qlik, ETL,
Scapple, JavaScript, TensorFlow,
PyTorch, CNN, Deep Learning,
Time Series Analysis, Predictive
Model, AI, Business Data
Analysis, Time-series analysis

Area Of Interests

Data Visualization, Data
Modeling, Data Reporting, Data
Manipulation, Data Extraction,
Data Collection, Data Mining,
Machine Learning, Data Cleaning,
Data-Driven Insights, Artificial
Intelligence, Predictive Analysis,
and statistical Analysis.

WORK EXPERIENCE

Senior Data Engineer, LTIMindtree, India

06/2022 – 07/2023 | India

- Developed efficient data pipelines with Azure Data Factory and automated ETL workflows using Python to streamline data processing and delta lake format and Azure Data Explorer or Kusto to work with the data.
- Established data pipelines and predictive models to increase clients' operational efficiency.
- Managed over 2TB of financial data in Snowflake, improving data warehousing efficiency by 25% and boosting processing speeds by 40%.
- Authored and optimized complex SQL queries for Oracle, reducing data retrieving by 30% while ensuring 99.9% accuracy for business reports.
- Created Power BI dashboards to facilitate decision-making using DAX logic to visualise findings, also Created dashboards and integrated HR procedures, leading to a 15% improvement in monthly KPI tracking and data transparency.
- Enhanced team productivity by teaching junior scientists statistical methods and process optimisation.

Data Analyst, IntelliCloud Apps Pvt. Ltd

01/2022 – 06/2022 | India

- Carried out the study of the industrial and real estate sectors, taking into account changes in the use of energy and production to inform investment choices.
- Boosted accuracy by 90%, utilizing statistics to identify trends in linear regression, cutting errors by 15%.
- Revolutionized property integration, enhanced client services, and achieved a 20% prediction error reduction.

Institute of Industrial and Systems Engineers at VIT, Research Analyst - Data Science

01/2020 – 12/2021 | India

- Developed and implemented a comprehensive data analysis pipeline for Google Play Store, analyzing a dataset of 10,842 entries to extract actionable insights and drive data-informed decisions.
- PySpark was used to facilitate data-driven decision-making through effective data translation, preprocessing, and large-scale analysis.
- Created powerful reports and visualisations that revealed important information about user behaviour and app performance.

PROJECTS

Covid Detection Using X-rays

- Utilising CNN, a deep learning model was developed to evaluate chest X-rays and precisely identify COVID-19 instances.
- Used the VGG-16 model and pre-processing techniques to classify COVID-19-positive and negative cases effectively.
- To enhance performance, the training dataset was supplemented with data augmentation techniques such as picture rotation and zoom.

Demographic and Socioeconomic Analysis of Remote Work: Trends, Impacts, and Visual Insights

- Identified significant workforce trends by examining preferences for remote work by gender, industry, and firm size.
- Investigated the effects of distant work on New York population growth and rental costs, providing information for urban planning and real estate.
- To assist organisations in making well-informed decisions, concise visual reports were created to illustrate important trends in remote work.

Insurance Management System

- Designed an ER diagram and used normalisation to improve the insurance management database's data integrity and cut down on redundancy.
- Developed triggers, indexing strategies, and stored procedures that reduced query execution time by 30% and automated policy modifications.
- Created Power BI dashboards for real-time policy and claims analytics and put role-based access control(RBAC) for safe data management.

CERTIFICATES

- Microsoft Certified: Azure Data Engineer Associate
- Microsoft Certified: Power BI Data Analyst Associate
- Python for data science (IBM certification)